

COMP1549 (2025/2026) - Advanced Programming

Security Access Control System - Demo Script

Demonstration Date: March 26, 2026

Environment: Windows 11, Java 17, Maven 3.9.14

C. Scenario Demonstration

1) Run the application with at least 3 different users:

- user1 (GUEST)
- user2 (STUDENT)
- user3 (STAFF)

2) Demonstrate access **ALLOWED**:

- GUEST → READ Printer (PUBLIC) → ALLOW
- STUDENT → READ Lecture Material (INTERNAL) → ALLOW
- STAFF → READ Lecture Material (INTERNAL) → ALLOW

3) Demonstrate **DENIED** access:

- GUEST → READ Exam Paper (CONFIDENTIAL) → REFUSE
- STUDENT → WRITE Exam Paper (CONFIDENTIAL) → REFUSE
- STAFF → READ Exam Paper (CONFIDENTIAL) → REFUSE

4) Show logged messages:

Each log includes: timestamp, userId, role, resource, operation, decision (ALLOW/REFUSE).

Example:

26-03-2026, user1, GUEST, Printer, READ, ALLOW

26-03-2026, user1, GUEST, Exam Paper, READ, REFUSE

D. Implementation Inspection

3) Most Challenging Design Pattern: Chain of Responsibility

This pattern is implemented in RuleChainAccessAuthorizer. It processes rules sequentially until a decision is found. If no rule applies, the default decision is REFUSE, ensuring secure access control.

4) Most Challenging JUnit Test:

The AccessControlPolicyTest uses parameterized testing to validate multiple scenarios across roles, resources, and operations. It ensures correctness of access rules and prevents security gaps.

Demo Checklist

- Environment verified
- Project builds successfully
- All tests pass
- Demo runs successfully
- 3 users demonstrated
- Allow and deny scenarios shown
- Logs displayed
- Design pattern explained
- JUnit test explained